6(iii)

$$re^{i\theta} = \frac{3+3i}{2+3i} + \frac{1+5i}{1-2i}$$

$$= -\frac{42}{65} + \frac{76}{65}i$$

$$= \sqrt{\frac{116}{65}}\; e^{i(0.661\pi)}$$

$$\theta = \pi - \tan^{-1}\left|\frac{76/65}{-42/65}\right|$$

$$= 118.93 \times \frac{\pi}{180}$$

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No.-07

(i) দে. প্র.

$$(a+ib)(c+id) = x+iy$$

বা, $ac + iad + ibc - bd = x+iy$

$\therefore (ac-bd) + i(ad+bc) = x+iy$ —— ①

① নং হতে উভয়পক্ষ সমীকৃত করে পাই,

$$\left.\begin{aligned} x &= ac-bd \\ y &= ad+bc \end{aligned}\right\} \begin{matrix} ② \\ ③ \end{matrix}$$

$L.S. = (a - ib)(c - id)$

$= ac - iad - ibc - bd$

$= (ac - bd) - i(ad + bc)$

$= x - iy$

$= R.S.$ //

(ii)

$$\frac{1 - ix}{1 + ix} = a - ib$$

বা, $\dfrac{1 + ix}{1 - ix} = \dfrac{1}{a - ib}$ [ধ-----]

বা, $\dfrac{1 + \cancel{ix} + 1 - \cancel{ix}}{\cancel{1} + ix - \cancel{1} + ix} = \dfrac{1 + a - ib}{1 - a + ib}$

বা, $\dfrac{\cancel{2}}{\cancel{2}ix} = \dfrac{1 + a - ib}{1 - a + ib}$

বা, $ix = \dfrac{1 - a + ib}{1 + a - ib}$

বা, $-x = \dfrac{i - ia - b}{1 + a - ib}$

গ, $x = \dfrac{b + ia - i}{1 + a - ib}$

$$= \frac{\{b + i(a-1)\}(1 + a + ib)}{(1 + a - ib)(1 + a + ib)}$$

$$= \frac{b + ab + ib^2 + i(a-1) + ia(a-1) - b(a-1)}{(1+a)^2 + b^2}$$

$$= \frac{b + \cancel{ab} + ib^2 + \cancel{ia} - i + ia^2 - \cancel{ia} - \cancel{ab} + b}{1 + 2a + a^2 + b^2}$$

$$= \frac{2b + i(a^2 + b^2) - i}{1 + 2a + (a^2 + b^2)}$$

$$= \frac{2b + i - i}{1 + 2a + 1}$$

$$\therefore x = \frac{2b}{2(a+1)} = \frac{b}{a+1}$$

4

$$\overline{z_1+z_2} = \overline{z_1}+\overline{z_2}$$

let, $z_1 = x_1+iy_1 \longrightarrow \overline{z_1} = x_1 - iy_1$

$z_2 = x_2+iy_2 \longrightarrow \overline{z_2} = x_2 - iy_2$

$$\overline{(x_1+x_2)-i(y_1+y_2)} \quad (2)$$

$\therefore z_1+z_2 = (x_1+x_2)+i(y_1+y_2)$

$\therefore \overline{z_1+z_2} = (x_1+x_2) - i(y_1+y_2) \quad (1)$

$z = (x+iy) \quad Re(z) = x$

$|z| = \sqrt{x^2+y^2} \quad Re(z) \leq |z|$

$$\boxed{\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}} \qquad |z_1 z_2| = |z_1||z_2|$$

(iv)

$$x+iy = \sqrt{\frac{p+iq}{r+is}} \quad \text{---} \ (1)$$

$$\therefore x-iy = \sqrt{\frac{p-iq}{r-is}} \quad \text{---} \ (2)$$

(1) × (2) ⇒

$$(x+iy)(x-iy) = \sqrt{\frac{p+iq}{r+is}} \cdot \sqrt{\frac{p-iq}{r-is}}$$

$$\therefore x^2+y^2 = \sqrt{\frac{(p+iq)(p-iq)}{(r+is)(r-is)}}$$

$$= \sqrt{\frac{p^2+q^2}{r^2+s^2}}$$

4

$|a^2+2| = -2$

$a =)$

$(x) = a + ib$

$(\ ) = 0$

$a^2+b^2 = 1$ //

//

7(vi)

$$x = \frac{2b + i(a^2+b^2) - i}{1 + 2a + a^2 + b^2}$$

$$x + i.0 = \frac{2b}{1+2a+a^2+b^2} + i\,\frac{a^2+b^2-1}{1+2a+a^2+b^2} \quad \text{——(1)}$$

$$\frac{a^2+b^2-1}{1+2a+a^2+b^2} = 0$$

$$\boxed{a^2+b^2=1}$$

$z = x + iy$

8 (iv) Ch. 2.

$z = x + iy$

$\therefore \bar{z} = x - iy$

$\therefore |z + i| = |\bar{z} + 2|$

বা, $|x + iy + i| = |x - iy + 2|$

বা, $|x + i(y + 1)| = |(x + 2) - iy|$

বা, $\sqrt{x^2 + (y+1)^2} = \sqrt{(x+2)^2 + y^2}$

বা, $x^2 + (y+1)^2 = (x+2)^2 + y^2$

বা, $\cancel{x^2} + \cancel{y^2} + 2y + 1 = \cancel{x^2} + 4x + 4 + \cancel{y^2}$

$$\boxed{\therefore 4x - 2y + 3 = 0}$$

ইহাই সঞ্চারপথ -

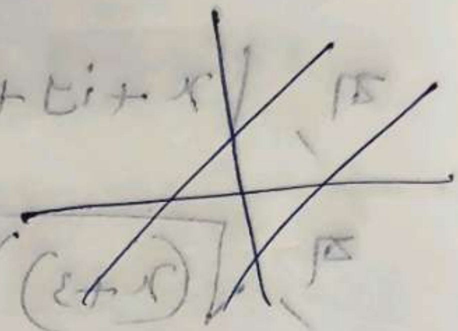
∴ ইহা - — — সঞ্চারপথের ।

$$ax^2+by^2+2hxy+2gx+2fy+c=0$$

$$ax^2+by^2=c$$

$$ax^2-by^2=1$$

$$\begin{vmatrix} a & h & g \\ h & b & e \\ g & e & f \end{vmatrix}$$



(iv)

$$Re\left(\frac{1}{z}\right)=1$$

বা, $Re\left(\frac{1}{x+iy}\right)=1$

$2(x+iy)$

বা, $Re\left(\frac{x-iy}{x^2+y^2}\right)=1$

বা, $Re\left(\frac{x}{x^2+y^2}-i\frac{y}{x^2+y^2}\right)=1$

বা, $\frac{x}{x^2+y^2}=1$

$\therefore x^2+y^2=x$

$\therefore x^2+y^2-x=0$

9

(iii) $z = x + iy$

$$(x-h)^2 + (y-k)^2 = r^2$$

$\therefore |z+3| = 4$

বা, $|x+iy+3| = 4$

বা, $\sqrt{(x+3)^2 + y^2} = 4$

$\therefore (x+3)^2 + y^2 = 4^2$ //

(ii)

$f(x) = x - 2$ $\quad (z = p + iq)$

$\therefore f(z+6) = z + 6 - 2$

$= z + 4$

$f(z-2) = z - 2 - 2$

$= z - 4$

$\therefore$

(vii)

$$\frac{z+i}{z+2}$$

$$= \frac{x+iy+i}{x+iy+2}$$

$$= \frac{x+i(y+1)}{(x+2)+iy}$$

$$= \frac{\{x+i(y+1)\}\{(x+2)-iy\}}{\{(x+2)+iy\}\{(x+2)-iy\}}$$

$$= \frac{x(x+2)-ixy+i(y+1)(x+2)+y(y+1)}{(x+2)^2+y^2}$$

$$= \frac{(x^2+2x+y^2+y)+i(\cancel{xy}+x+2y+2-\cancel{xy})}{(x+2)^2+y^2}$$

$$= \frac{(x^2+y^2+2x+y)+i(x+2y+2)}{(x+2)^2+y^2} \quad (1)$$

① যা' সম্পূর্ণ কাল্পনিক হলে, বাস্তব অংশ শূন্য হবে,

$$\therefore \frac{x^2+y^2+2x+y}{(x+2)^2+y^2}=0$$

$$\therefore x^2+y^2+2x+y=0$$

10 (iv)

$$x+iy = 2e^{-i\theta} \quad ①$$

$$z = re^{i\theta}$$

$$\bar{z} = re^{-i\theta}$$

$$\therefore x-iy = 2e^{+i\theta} \quad ②$$

① × ②,

$$x^2+y^2 = 4e^{-i\theta+i\theta}$$

$$= 4e^0 = 4$$

4(iii)

$2n\pi+(\;)$

z_1, z_2

$z_1+z_2=4$ —(1)

$z_1 z_2 = 8$

$(z_1-z_2)^2 = (z_1+z_2)^2 - 4z_1z_2$

$= 4^2 - 4\cdot 8$

$= 16 - 32$

$(z_1-z_2)^2 = -16$

$\sqrt{-16}$

$z_1 - z_2 = 4i$ —(2)

$\sqrt{16}\times\sqrt{-1}$

$4i$

(1)+(2) =>

$2z_1 = 4+4i$

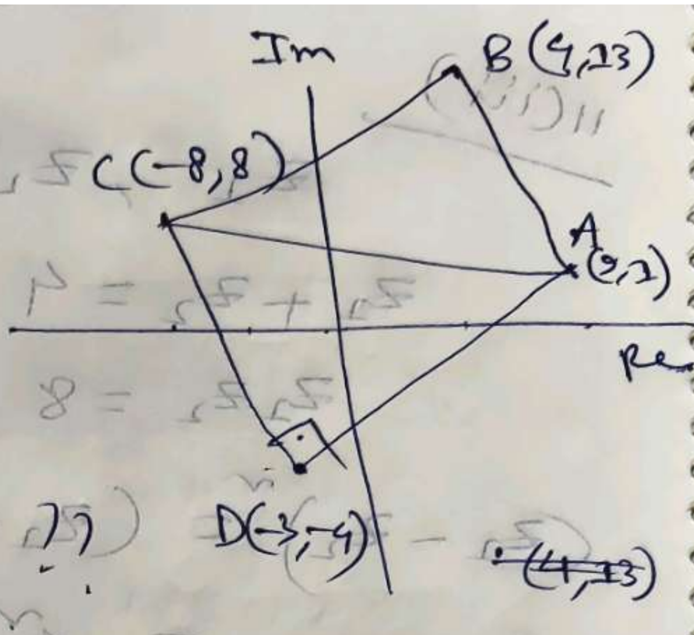
$\therefore z_1 = 2+2i$

$x-4i \rightarrow -3+iy$

$x+4i$

$z_2 = ?$

12



$AB = BC = CD = DA$

$AC = ?$

$AD^2 + DC^2 = AC^2$

3.6.1 ধরি,

$\sqrt{a+ib} = x+iy$

বা, $a+ib = x^2 + i2xy - y^2$

$\therefore a = x^2 - y^2$

$\therefore b = 2xy$

$x^2 - y^2 = a$ —(1)

$(x^2+y^2)^2 = (x^2-y^2)^2 + 4x^2y^2$

$= a^2 + b^2$

$\therefore x^2+y^2 = \sqrt{a^2+b^2}$ —(2)

① + ② ⟹

$$2x^2 = a + \sqrt{a^2+b^2}$$

$$x^2 = \frac{1}{2}\left(a + \sqrt{a^2+b^2}\right)$$

$$\boxed{\therefore x = \pm \frac{1}{\sqrt{2}}\left(a + \sqrt{a^2+b^2}\right)^{1/2}}$$

$$\boxed{\therefore y = \pm \frac{1}{\sqrt{2}}\left(a + \sqrt{a^2+b^2}\right)^{1/2}}$$

$$\sqrt{24+3i} = ? \quad (x+iy)$$

$$\therefore x = \pm \frac{1}{\sqrt{2}}\left(24 + \sqrt{24^2+3^2}\right)^{1/2}$$

$y = ?$

3(B)

1(iv)

$$\sqrt{-8-6\sqrt{-1}} = \sqrt{-8-6i}$$

ধরি, $\sqrt{-8-6i} = x + iy$

Way-2

$$x = \pm\frac{1}{\sqrt{2}}\left\{-8+\sqrt{64+36}\right\}^{1/2}$$

$$= \pm\frac{1}{\sqrt{2}}\left\{-8+10\right\}^{1/2}$$

$$= \pm\frac{1}{\sqrt{2}}\sqrt{2} = \pm 1$$

$$y = \pm\frac{1}{\sqrt{2}}\left\{8+10\right\}^{1/2}$$

$$= \sqrt{9 \times 2}$$

$$= \pm\frac{1}{\sqrt{2}}\sqrt{18} = \pm\frac{1}{\sqrt{2}}3\sqrt{2}$$

$$= \pm 3$$

$\therefore \sqrt{-8-6i} = \cancel{\pm 1 \pm 3i}$

$= \pm(1 - 3i)$

Way-3

$\cancel{\sqrt{-8-6i} = \sqrt{-3^2 - 2\cdot 3\cdot i + i^2}}$

(ii)

$\cancel{\sqrt{-3-4i} = \sqrt{-2^2 - 2\cdot 2\cdot i + 1^2}}$

$\cancel{\sqrt{-(2+2\sqrt{3}\,i)} = \sqrt{(\sqrt{3})^2 + 2(\sqrt{3})i}}$

$\cancel{= \sqrt{-((\sqrt{3})^2 + 2\cdot\sqrt{3}\cdot i - 1)}}$

$\sqrt{-2-2\sqrt{3}i} = \cancel{\sqrt{(-\sqrt{3})^2 + 2(-\sqrt{3})i - 5}}$

$\sqrt{1 - 2\cdot 1\cdot \sqrt{3}i + (\sqrt{3}i)^2}$

$= \sqrt{(1-\sqrt{3}i)^2}$

$= \pm(1-\sqrt{3}i)$

প্রশ্ন: দেখাও, দেখাও প্রমাণ,

$\sqrt{a+ib} = x+iy$ হলে,

$$x = \pm\frac{1}{\sqrt{2}}\left\{a+\sqrt{a^2+b^2}\right\}^{1/2}$$

$$y = \pm\frac{1}{\sqrt{2}}\left\{-a+\sqrt{a^2+b^2}\right\}^{1/2}$$

উদা:।

$\sqrt{-8-6i} = x+iy$